

QUANTUM VERSUS CLASSICAL COMPUTERS

1.) What might a physical realization of a quantum computer look like?

Solution: There are many possibilities. It must be a two-state system. For this reason, spin-1/2 particles in optical traps have been considered, as have tunneling electrons in current-biased Josephson junctions (superconductor-insulator-superconductor interfaces), hyperfine levels in atoms, polarization states of photons.

2.) Tell your partner one important advantage of a quantum computer over a classical computer.

Solution: It can solve certain problems more efficiently than classical computers. The case with the greatest impact on present day society would be Shor's algorithm, which shows that a quantum computer could factor prime numbers in polynomial time. Since prime factors are used in encryption, this would allow quantum computers to break most modern security. There are other examples of things quantum computers can do better than classical computers, such as Deutsch's Algorithm. But these other examples are fairly contrived. Grover's search algorithm allows a quantum computer to search an unsorted database of N entries in a time proportional to $\sqrt{N}$, compared to a classical computer that requires a time that is proportional to N .

3.) Are there any disadvantages to a quantum computer over a classical computer? Can we overcome them?

Solution: Yes! Measurements are probabilistic! They would be / are very noisy, easily altered by the slightest cosmic ray, etc. In some cases, we can get around this. For example, in the case of prime factoring, one would simply check the result, since multiplying primes is easy. From McIntyre (page 517): "Likewise, as hard as it is to find a needle in a haystack, it is simple to determine if the object you find is a needle."

4.) Your friend says, "So I read that a state in a system of N qubits (in a quantum computer) can store 2^N pieces of information. And that's supposed to be a great advantage. But... a set of N bits (plain old bits from the 1960's) also has 2^N possible states. So... what is the point of a quantum computer?" How do you respond?

Solution: There was a subtle difference in wording. The quantum computer can store 2^N pieces of information (the coefficients) in a single state, and even use those pieces of information in a calculation. The classical computer could be in any of 2^N possible states, but in that state it only has N bits of information. For a large number of qubits, $2^N \gg N$.

5.) Consider a general state of a 3-qubit system. By writing out explicit states, find how many pieces of information can be contained in a state in such a system if the state is (a) a product state, and (b) an entangled state.

Solution: (a) A product state can be written as $|\psi\rangle_{\text{product}} = (a_0|0\rangle_A + a_1|1\rangle_A) \otimes (b_0|0\rangle_B + b_1|1\rangle_B) \otimes (c_0|0\rangle_C + c_1|1\rangle_C)$. For each particle/qubit, one coefficient can be chosen freely, and the other must be chosen such that the probability of finding that particular particle in some state is one. Thus, for each of the N qubits, we have one free coefficient. We can store 3 pieces of information in this state.

(b) The entangled state can be written as $|\psi\rangle_{\text{entangled}} = \alpha_1|0\rangle_A|0\rangle_B|0\rangle_C + \alpha_2|1\rangle_A|0\rangle_B|0\rangle_C + \alpha_3|0\rangle_A|1\rangle_B|0\rangle_C + \alpha_4|0\rangle_A|0\rangle_B|1\rangle_C +$

$$\alpha_5|1\rangle_A|1\rangle_B|0\rangle_C + \alpha_6|1\rangle_A|0\rangle_B|1\rangle_C + \alpha_7|0\rangle_A|1\rangle_B|1\rangle_C + \alpha_8|1\rangle_A|1\rangle_B|1\rangle_C$$

Clearly, there are 8 coefficients. Choosing 7 of them determines the 8th because of normalization. So, we have $2^N - 1 = 7$ pieces of information we can store in this state.

6.) Suppose you have a system of N spin-1/2 particles. They are nearby, but not entangled with each other, meaning you can write their overall state as $|\psi\rangle_{\text{product}} = (a_0|0\rangle_A + a_1|1\rangle_A) \otimes (b_0|0\rangle_B + b_1|1\rangle_B) \otimes (c_0|0\rangle_C + c_1|1\rangle_C) \dots$ (a) If you measure the state of the system, how many states are possible? (b) How many coefficients are required to completely specify a state? (c) Is this the same, or different from a classical computer with N bits?

Solution: (a) 2^N states are possible. 2 states for each of the N qubits. (b) N coefficients are needed. For each qubit, we can pick/specify one of the coefficients, and the second is determined by normalization. (c) The number of possible states, and the number of units of information are both the same as in a classical computer.

- 7.) Suppose you have a system of N spin-1/2 particles. They are entangled so that the state in which one particle is measured is related to the state in which another particle will be measured. In this case, we can write the overall state as $|\psi\rangle_{\text{entangled}} = \alpha|0\rangle_A|0\rangle_B|0\rangle_C\cdots + \beta|1\rangle_A|0\rangle_B|0\rangle_C\cdots + \gamma|0\rangle_A|1\rangle_B|0\rangle_C\cdots + \dots$ (a) If you measure the state of the system, how many states are possible? (b) How many coefficients are required to completely specify a state? (c) Is this the same or different from a classical computer with N bits?

Solution: (a) Still 2^N states are possible. (b) Now there are $2^N - 1 \approx 2^N$ coefficients that can be freely chosen. (c) No! This is much more data than can be stored/manipulated by a classical computer with an equivalent number of bits.

- 8.) Given the results of some of the previous questions, would you say the point of a quantum computer is more (i) to make classical measurements to collapse the system into certain states, or (ii) to explicitly avoid classical measurements long enough to entangle states and use the properties of their entanglement?

Solution: The point is definitely to use entanglement. The previous questions have shown that making classical measurements, forcing states into eigenstates as opposed to entanglements reduces the quantum computer to a classical one.

OPERATORS AND MEASUREMENT

- 9.) Much of our work in quantum mechanics (especially in undergrad) focuses on Hermitian operators. Why was that? What property did they have and why did we need it?

Solution: We needed Hermitian operators because they have real eigenvalues. If we want them to correspond to physical observables, then we need a measurement associated with that operator to yield a real number.

- 10.) Now, if we are not making classical measurements, we can still transform a qubit via some physical process (this is called a quantum gate). Mathematically, we represent such physical processes using operators. What properties should these operators have?

Solution: Now that we are avoiding measurements, the focus will not be so heavily on Hermitian operators. Now, we are allowing systems to simply time-evolve in the presence of a certain type of “gate”. $\hat{U} = e^{-i\hat{H}t/\hbar}$ is unitary... always, provided that the Hamiltonian is still Hermitian. We may also simply “act with operators”, which is not the same as making a measurement. If we use operators, they must do two things: they must alter the coefficients without altering the coherence of the state, and they must preserve the Euclidean norm; that is, such operators must be unitary.

- 11.) If systems are simply evolving in time under the influence of certain Hamiltonians that we engineer, the time-evolution is described by the propagator $\hat{U} = e^{-i\hat{H}t/\hbar}$. What kind of operator is this?

Solution: This is unitary, always, provided the Hamiltonian is still Hermitian.

- 12.) Consider some Hermitian operator $\hat{O}$, which is not unitary, but which is related to a unitary operator through $\hat{O} = c\hat{U}$. Let $|0\rangle, |1\rangle$ be eigenstates of $\hat{O}$ and $\hat{U}$ with eigenvalues o_0, o_1 , and u_0, u_1 . Note that since $\hat{U}$ is unitary, $|u_0|^2 + |u_1|^2 = 1$. Let the initial state be $|\psi_i\rangle = \alpha|0\rangle + \beta|1\rangle$.

- (a) Is acting with an operator the same thing as making a measurement of the physical property associated with the operator?

- (b) What final state results from acting with $\hat{O}|\psi_i\rangle$? Is this a reasonable state? Why or why not?
 (c) What final state results from acting with $\hat{U}|\psi_i\rangle$? Is this a reasonable state? Why or why not?
 (d) Explain to your partner why we often hear that quantum logic gates are unitary operators.

Solution:

(a) NO! Making a measurement collapses the state into either $|0\rangle$ or $|1\rangle$. Acting with the operator distributes through to the terms in a mixed state.

(b) $\hat{O}|\psi_i\rangle = \hat{O}(\alpha|0\rangle + \beta|1\rangle) = \alpha o_0|0\rangle + \beta o_1|1\rangle$. But wait! That is no longer normalized, for general o_0, o_1 !

(c) $\hat{U}|\psi_i\rangle = \hat{U}(\alpha|0\rangle + \beta|1\rangle) = \alpha u_0|0\rangle + \beta u_1|1\rangle$. This is a reasonable state, since $\langle\psi_i|\hat{U}^\dagger\hat{U}|\psi_i\rangle = \langle\psi_i|\psi_i\rangle = 1$

(d) Avoiding measurement means we don't need all our operators to be Hermitian any more. Acting with an operator on a state will make the state un-normalized unless the operator is Unitary. Time evolution in the presence of a Hamiltonian we control is unitary anyway. We will use unitary matrices for quantum logic gates.

ENTANGLEMENT

13.) (from R.V.A at Bluffton) Imagine a 2-qubit entangled system: $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Alice and Bob share custody of the state. Alice takes one qubit and Bob takes the other.

- (a) What is the probability that Alice measures her qubit to be 0?
 (b) What state is the state of the qubit system after Alice measures her qubit to be 0?
 (c) What if Bob measures his qubit before Alice? What are the possible outcomes and probabilities for Bob's measurement?
 (d) What if Bob measures his qubit after Alice measures a 0 on hers? What are the possible outcomes and probabilities for Bob's measurement?

Solution:

- (a) 50%
 (b) $|00\rangle$
 (c) 50% probability of measuring $|0\rangle$ and 50% probability of measuring $|1\rangle$
 (d) 100% probability of measuring $|0\rangle$

14.)(from K.S at UC Boulder) Here is a 2-qubit state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|11\rangle$$

- (a) Suppose you measure the first qubit and obtain a measurement of $|0\rangle$. What is the state after this measurement?
 (b) If you now measure the second qubit, what are the possible results and probabilities of those results?
 (c) Supposed instead you measure the first qubit and obtain a measurement of $|1\rangle$. What is the state after this measurement?
 (d) If you now measure the second qubit, what are the possible results and probabilities of those results?
 (e) Is this an entangled state? Justify.

Solution:

- (a) The state $|11\rangle$ drops off, and the new state is a linear combination of $|00\rangle$ and $|01\rangle$. This new state needs to be normalized. After normalizing by the length, the new state is: $\frac{2}{\sqrt{3}}\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle\right) = \frac{\sqrt{2}}{\sqrt{3}}|00\rangle + \frac{1}{\sqrt{3}}|01\rangle$.
- (b) We have probability of 2/3 that we obtain $|00\rangle$ and probability of 1/3 that we obtain $|01\rangle$. Fortunately, these add up to 1.
- (c) The state $|11\rangle$ is the only one surviving, and we obtain, after normalizing: $|11\rangle$.
- (d) We have probability 1 that we obtain $|11\rangle$.
- (e) Yes. We get two different outcomes to measuring the first qubit then differing probabilities for the outcome measuring the second. Hence, the qubits are not independent.

VON NEUMANN ENTROPY

15.) The von Neumann entropy is defined as $S(\rho) = -\text{Tr}[\rho \log \rho]$.

- (a) Prove that for any two positive definite matrices (i.e., matrices with positive, non-zero eigenvalues) A and B we have $\log(A \otimes B) = \log(A) \otimes \mathbb{I} + \mathbb{I} \otimes \log(B)$
- (b) How can we calculate the logarithm of a matrix if (at least) one of its eigenvalues is zero?
- (c) Write the von Neumann entropy of ρ in terms of its eigenvalues and eigenvectors.

Solution:

- (a) We first address the case $A, B > 0$. Let A and B have spectral decompositions: $A = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$, $B = \sum_j \mu_j |\phi_j\rangle\langle\phi_j|$, where the eigenvalues by the positive definiteness assumption satisfy $\lambda_i > 0$ and $\mu_j > 0$ for all i, j . In particular $\log(\lambda_i \mu_j)$, $\log(\lambda_i)$, and $\log(\mu_j)$ are well-defined for all i and j . Then, the spectral decomposition of $A \otimes B$ is: $A \otimes B = \sum_{ij} \lambda_i \mu_j |\psi_i\rangle\langle\psi_i| \otimes |\phi_j\rangle\langle\phi_j|$

Now we can compute $\log(A \otimes B)$:

$$\begin{aligned}
 \log(A \otimes B) &= \log\left(\sum_{ij} \lambda_i \mu_j |\psi_i\rangle\langle\psi_i| \otimes |\phi_j\rangle\langle\phi_j|\right) \\
 &= \sum_{ij} (\log(\lambda_i) + \log(\mu_j)) |\psi_i\rangle\langle\psi_i| \otimes |\phi_j\rangle\langle\phi_j| \\
 &= \left(\sum_i \log(\lambda_i) |\psi_i\rangle\langle\psi_i|\right) \otimes \left(\sum_j |\phi_j\rangle\langle\phi_j|\right) + \left(\sum_i |\psi_i\rangle\langle\psi_i|\right) \otimes \left(\sum_j \log(\mu_j) |\phi_j\rangle\langle\phi_j|\right) \\
 &= \log(A) \otimes \mathbb{I} + \mathbb{I} \otimes \log(B)
 \end{aligned}$$

The last step used that $\sum_i |\psi_i\rangle\langle\psi_i| = \mathbb{I}$ and $\sum_j |\phi_j\rangle\langle\phi_j| = \mathbb{I}$.

- (b) Note on positive definite matrices. Now for the case $A, B \geq 0$. If A and B have 0 as an eigenvalue, then so does $A \otimes B$. However, by definition of the matrix logarithm, we apply the logarithm only to the non-zero eigenvalues occurring in the spectral decompositions and leave the 0-eigenvalue untouched (since the corresponding summands in the spectral decomposition anyways vanish). Hence, we don't run into any issues about the logarithm not being well-defined. This can be made more formal when written in terms of eigenprojections.
- (c) Diagonalize ρ to obtain $S(\rho) = -\sum_x \lambda_x \log \lambda_x$, where λ_x are the eigenvalues of ρ .